



THREAT ADVISORY

**ATTACK
REPORT**

Lazarus distributes Nukesped to VMware Horizon Servers by exploiting Log4J

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24 May 2022

Admiralty code

A2

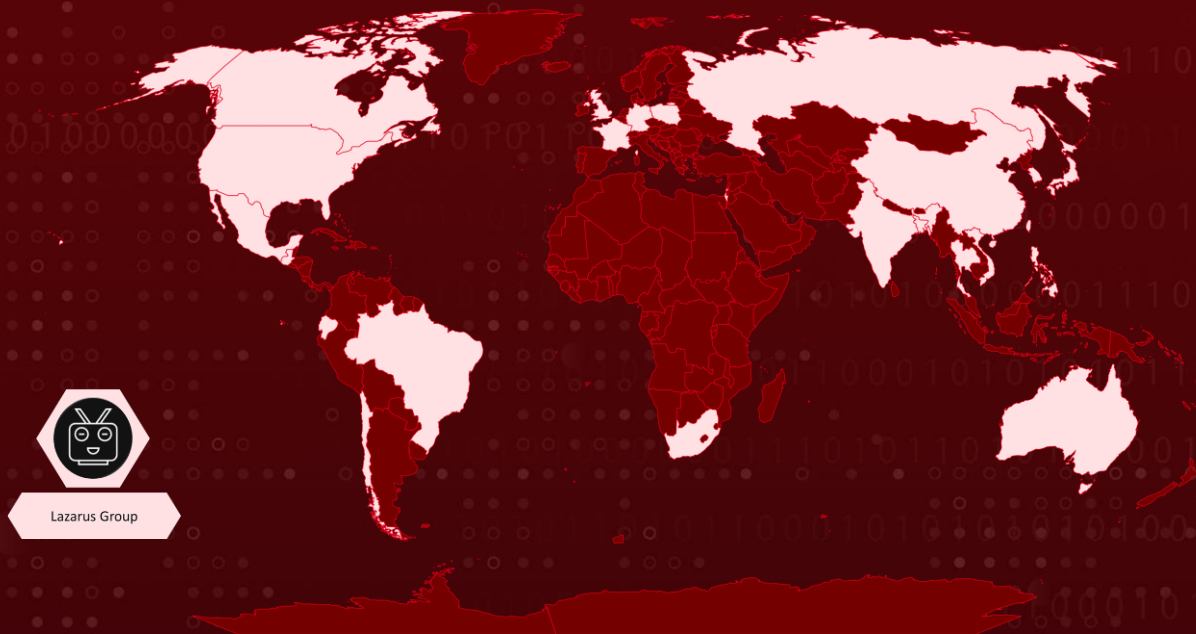
TA Number

TA2022107

Summary

Lazarus, a North Korean threat actor group, is deploying Nukesped (aka Manuscript) malware on unpatched VMware Horizon servers by exploiting the Log4j remote code execution vulnerability.

Actor Map



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CVE

CVE	NAME	PATCH
CVE-2021-44228	Apache Log4j Vulnerability	

Potential MITRE ATT&CK TTPs

TA0001 Initial Access	TA0002 Execution	TA0003 Persistence	TA0004 Privilege Escalation
TA0005 Defense Evasion	TA0011 Command and Control	TA0007 Discovery	TA0009 Collection
TA0042 Resource Development	TA0006 Credential Access	T1566 Phishing	T1566.001 Phishing: Spearphishing Attachment
T1587 Develop Capabilities	T1587.003 Develop Capabilities: Digital Certificates	T1204 User Execution	T1204.002 User Execution: Malicious File
T1547 Boot or Logon Autostart Execution	T1547.001 Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder	T1543 Create or Modify System Process	T1543.003 Create or Modify System Process: Windows Service
T1140 Deobfuscate/Decode Files or Information	T1036 Masquerading	T1036.005 Masquerading: Match Legitimate Name or Location	T1112 Modify Registry
T1027 Obfuscated Files or Information	T1083 File and Directory Discovery	T1005 Data from Local System	T1105 Ingress Tool Transfer
T1573 Encrypted Channel	T1573.001 Encrypted Channel: Symmetric Cryptography	T1059 Command and Scripting Interpreter	T1059.001 Command and Scripting Interpreter: PowerShell
T1056 Input Capture	T1056.001 Input Capture: Keylogging	T1113 Screen Capture	T1059.001 Command and Scripting Interpreter: PowerShell

Technical Details

#1

Lazarus initiates the **attack** by exploiting the **Log4j** vulnerability via VMware Horizon's Apache **Tomcat** server to execute a **PowerShell** command. This **PowerShell** command will eventually result in the installation of the **NukeSped backdoor** on the server.

#2

NukeSped is a **backdoor** malware that can receive and execute **threat actors'** commands from the **C&C** server. It communicates securely with the C2 infrastructure using **RC4 encryption**. Depending on the command, **NukeSped** can perform **keylogging**, take **screenshots**, and **file** and **shell** tasks. The variant used in this attack is developed in **C++**.

#3

Lazarus deploys **NukeSped** to install additional console-based **information-stealer** malware that steals data from web browsers. Later, the **information** gathered can be utilized in **lateral movement** attacks.



Actor Detail

NAME	ORIGIN	MOTIVE	TARGET LOCATIONS	TARGET INDUSTRIES
Lazarus Group (Labyrinth Chollima, Group 77, Hastati Group, Whois Hacking Team, NewRomanic Cyber Army Team, Zinc, Hidden Cobra, Appleworm, APT-C-26, ATK 3, SectorA01, ITG03)	North Korea	Information theft and espionage, Sabotage and destruction, Financial crime	Australia, Bangladesh, Brazil, Canada, Chile, China, Ecuador, France, Germany, Guatemala, Hong Kong, India, Israel, Japan, Mexico, Philippines, Poland, Russia, South Africa, South Korea, Taiwan, Thailand, UK, USA, Vietnam	Aerospace, Defense, Engineering, Financial, Government, Media, Shipping and Logistics, Technology and Cryptocurrencies

Vulnerability Details

CVE ID	AFFECTED PRODUCTS	AFFECTED CPE	CWE ID
CVE-2021-44228	VMware Horizon versions 8.x, 7.x	cpe:2.3:a:vmware:vmware_horizon:*:*:*:*:*:*:*	CWE-94

Indicator of Compromise (IOC)

TYPE	VALUE
MD5	87a6bda486554ab16c82bdfb12452e8b, 830bc975a04ab0f62bfedf27f7aca673, 131fc4375971af391b459de33f81c253, 827103a6b6185191fd5618b7e82da292, 1875f6a68f70bee316c8a6eda9ebf8de, 85995257ac07ae5a6b4a86758a2283d7, 47791bf9e017e3001ddc68a7351ca2d6
URLs	hxxp://185.29.8[.]18/htroy.exe, mail.usengineergroup[.]com:8443
IPv4:Port	185.29.8[.]18:8888, 84.38.133[.]145:443, 84.38.133[.]16:8443
File Name	svc.exe, srvCredit.exe, runhostw.exe, javarw.exe

Patch Links

<https://www.vmware.com/security/advisories/VMSA-2021-0028.html>

References

<https://asec.ahnlab.com/en/34461/>

What Next?

Book a free demo with **HivePro Uni5** to check your exposure to this advisory. HivePro Uni5 is a Threat Exposure Management Solution that proactively reduces an organization's attack surface before it gets exploited.



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